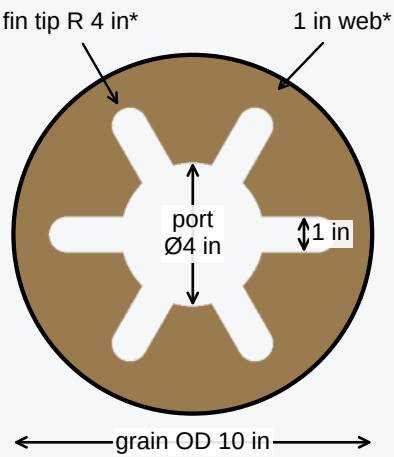


Finocyl grain burnback and internal ballistics

10 in OD × 60 in · Ø4 in port · 6 fins × 1 in · AP/HTPB/Al composite (68/18/14)

1 Problem

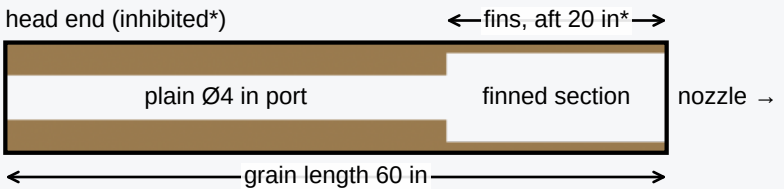


Find burn area vs web burned, chamber pressure and thrust vs time for a finocyl grain, and show that the answer does not depend on the mesh.

Grain 10 in OD × 60 in long, Ø4 in port, 6* fins × 1 in wide with semicircular tips*, OD case-bonded, both ends inhibited*

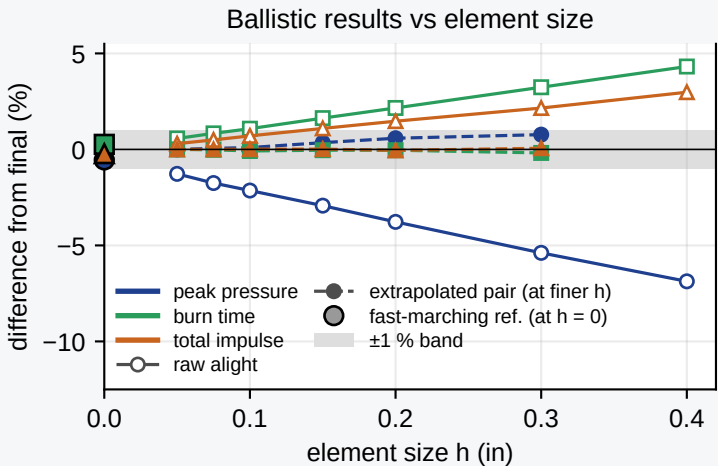
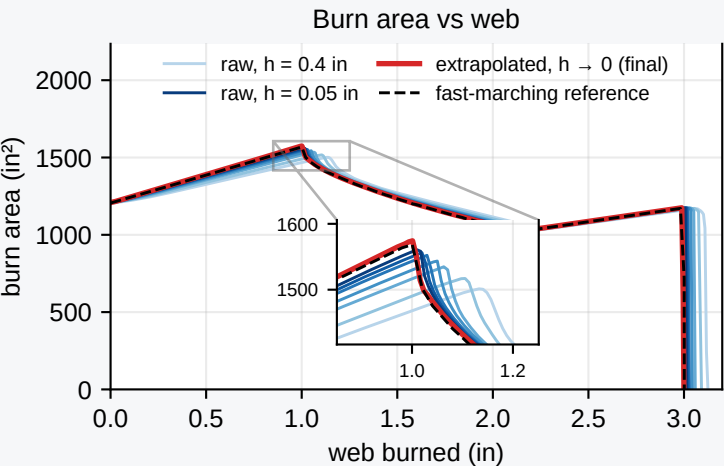
Propellant AP / HTPB / Al composite (68 / 18 / 14 %)
 $\rho = 1800 \text{ kg/m}^3$, $c^* = 1580 \text{ m/s}$, $\gamma = 1.14$
 $r = 0.35 \text{ in/s}$ at 1000 psi, $n = 0.35$

Nozzle throat Ø2.70 in (sized for 1000 psi peak), expansion ratio 8, sea level



* Not specified in the request: assumed, adjustable input.
Propellant values are representative handbook-level numbers, not a specific qualified formulation.

3 Mesh sensitivity



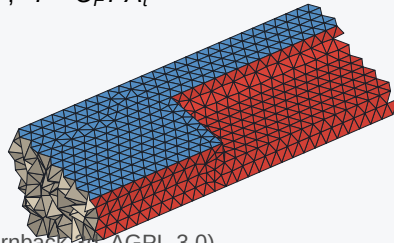
- Raw alight results drift with the mesh: peak pressure spans 5.6 % over $h = 0.4$ to 0.05 in and still moves 0.49 % at the last refinement. The error falls linearly with h (first order).
- Extrapolating two meshes removes it: for every pair from $0.2/0.15$ in down, peak pressure, burn time and total impulse stay within 0.35 %, and the last two pairs differ by at most 0.05 %.
- The final result differs from the independent fast-marching reference by 0.55 % in peak pressure, 0.26 % in burn time and 0.22 % in total impulse. CFL and convergence tolerance do not change the converged field.
- Reference case source: computed in this study, not taken from literature. scikit-fmm 2025.6.23 marches the distance from the initial port, voxelised from the grain definition on a 0.035 in Cartesian grid (37 M nodes, one quadrant); scikit-image 0.26.0 marching cubes gives the burn area. The closed-form distance on the same grid agrees with it within 0.16 % in peak pressure.

2 Approach

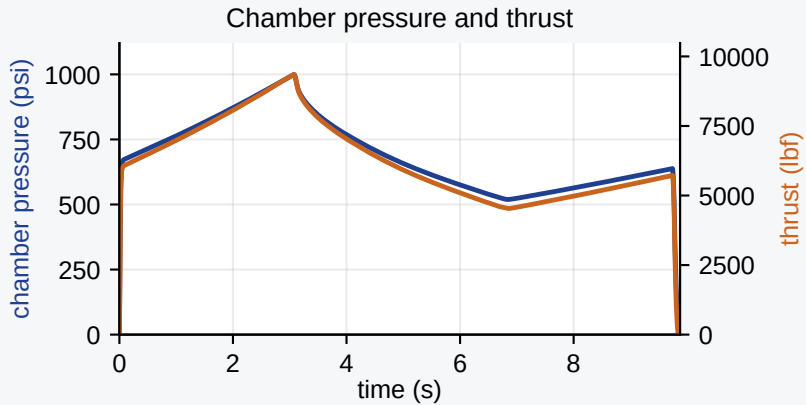
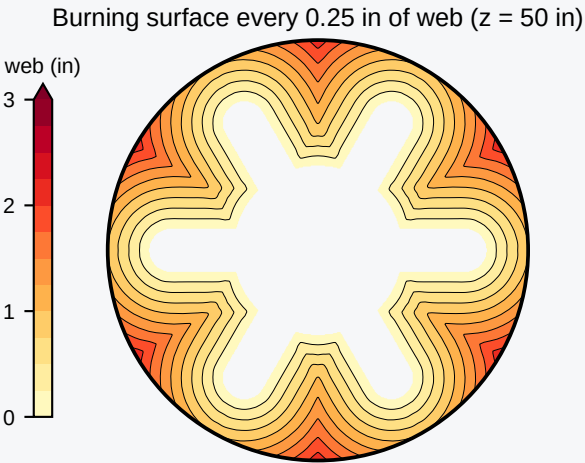
- CAD** build123d parametric solid → STEP; one 1/12 symmetry wedge (half a fin).
- Mesh** Gmsh tetrahedra. Boundaries: inlet = port and fin faces, outlet = case wall and end faces, symmetry = wedge planes.
- Burnback** alight solves $|\nabla u| = 1$, where u is the web burned when the flame arrives (the burnback-3d kernels behind a command line).
- Burn area** $A(w)$ = area of the surface $u = w$, by marching tetrahedra, × 12.
- Mesh error** The solver is first-order accurate (numerical diffusion), so two meshes are Richardson-extrapolated to zero element size (final pair: 0.075 and 0.05 in).
- Check** Independent fast-marching distance field (scikit-fmm) with marching-cubes surfaces (scikit-image) on a Cartesian grid.
- Ballistics** 0-D chamber: $dP/dt = (RT/V)[\rho A(w)r - PA_t/c^*]$, $r = aP^n$, $F = C_F PA_t$
 - inlet: initial burning surface
 - outlet: case wall (and inhibited ends)
 - symmetry plane
 - cut through the mesh (display only)

8 in slice of the solved wedge at the fin start,
0.3 in elements shown (finest mesh: 9.8 M tetrahedra)

Solver: alight 0.1.0, github.com/jakeboening/alight, a fork of burnback-3d (codeberg.org/iff/burnback-3d, AGPL-3.0).
Not modelled: erosive burning, axial pressure drop, throat erosion, ignition transient.



4 Results



Peak / mean pressure	1000 / 689 psi
Peak thrust	9,355 lbf
Burn time (10 % to 10 %)	9.82 s
Total impulse	61,559 lbf·s (274 kN·s)
Specific impulse, sea level	254 s
Propellant mass	110.0 kg (242 lbm)
Burn area, initial / peak	1,208 / 1,574 in²
Web at fin burn-through / burnout	1.00 / 3.00 in

Burn area peaks when the fin tips reach the case wall, falls as the fin slots burn out, then rises again in the plain port until burnout.